

Phase 6 - README

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PHASE 6 — Financial Intelligence

Real Estate AI Decision Intelligence Platform

Follows Phase 5 in the Master A-Z Prompt’s 10-phase architecture. Matches the established code-structure convention from Phases 1/3/4/5 exactly: live PostgreSQL via `db/connection.py`, `src/<topic>/*.py` modules, three orchestrators (Markdown report, numbered CSV export, chart PNGs), and an automated test suite.

What was built

Module	Purpose	Master Prompt Section
<code>data_loader.py</code>	Sources every OBSERVED input (price, rent, expenses, vacancy, locality appreciation) from Phase 1 tables. Flags unreliable locality appreciation (inherited Phase 3 data bug).	6.2
<code>engine.py</code>	Deterministic cash-flow schedule, NOI, yields, cap rate, ROI, IRR, NPV, DSCR, cash-on-cash, payback, terminal CAGR. Implements the Missing Input Policy.	6.3, 6.4, 6.7
<code>scenarios.py</code>	Base / Upside / Downside + 6-driver sensitivity analysis. Reuses <code>engine.py</code> — no duplicated math.	6.5, 6.6
<code>batch_runner.py</code>	Reproducible demo sample selection + batch execution.	—
<code>export_results_csv.py</code>	6 numbered CSVs → <code>docs/financial_results_csv/</code> .	6.8
<code>generate_charts.py</code>	4 charts → <code>docs/finance_charts/</code> .	—
<code>test_phase6.py</code>	25 automated tests (all passing, includes 2 post-delivery audit-fix regression tests).	6.9

Full detail: `docs/phase6_report.md` (phase completion report), `docs/limitations.md`.

Notable finding

Cross-checking historical price CAGR for all 90 localities found 13 (14.4%) with CAGRs above 30%/yr — up

to 132.75%/yr — inherited from the Phase 3-documented data-generator compounding bug. The engine now refuses to use these as a default `appreciation_rate` and returns `INSUFFICIENT DATA` for that field instead, unless a caller explicitly overrides it. Details in `docs/phase6_report.md` Section 6.

How to run

With a live database (production path)

```
cd phase6
pip install -r requirements-phase6.txt --break-system-packages
export DB_USER=...
export DB_PASSWORD=...

python3 src/finance/export_results_csv.py      # docs/financial_results_csv/*.csv
python3 src/finance/generate_charts.py        # docs/finance_charts/*.png
```

Local re-check (no database — what was used to produce this deliverable)

```
cd phase6
pip install -r requirements-phase6.txt --break-system-packages

python3 src/finance/export_results_csv.py --source csv
python3 src/finance/generate_charts.py --source csv

python3 -m pytest src/finance/test_phase6.py -q
```

Single-property example

```
from src.finance.data_loader import load_property_financials_from_csv
from src.finance.engine import FinancialAssumptions, FinancingTerms, run_financial_engine
from src.finance.scenarios import run_scenarios, run_sensitivity

observed = load_property_financials_from_csv(1)
assumptions = FinancialAssumptions(
    property_id=1,
    annual_operating_expenses=180000, # supply if this property has no expense history
    vacancy_rate=0.05,
    appreciation_rate=0.08,
    financing=FinancingTerms(down_payment_pct=0.20, loan_interest_rate=0.09, loan_term_years=20),
)
result = run_financial_engine(assumptions, observed)
scenarios = run_scenarios(assumptions, observed)
sensitivity = run_sensitivity(assumptions, observed)
```